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Mushroom Farming

Saurab Dulal, John Thompson
{sdulal, jrthmpsn}@memphis.edu

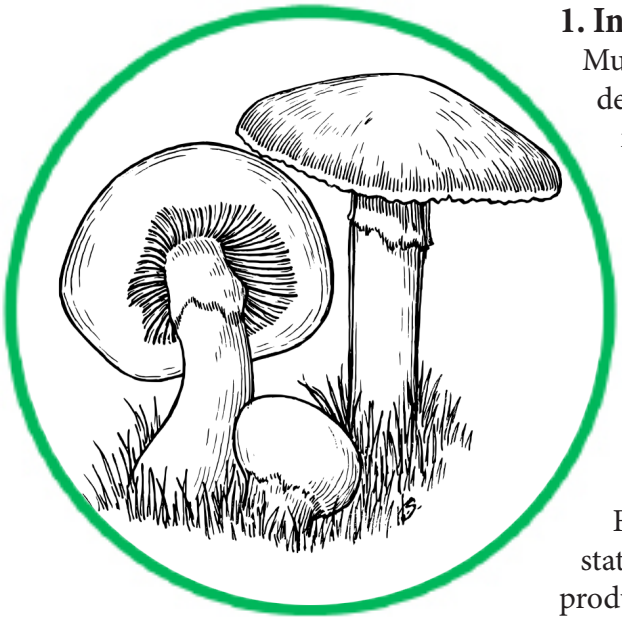

Fig 1: Oyster Mushroom

1. Introduction

Mushroom is a fungus shaped like an umbrella that grows in decomposing organic matter. Unlike most plants that produce seeds, mushrooms produce spores to continue the reproduction process.

Mushroom farming is the process encompassing the growth, harvest, storage, and selling of mushrooms. It is scientifically known as floriculture.

There are thousands of types of mushrooms and they require little to thrive. Because of their resiliency, mushroom farming is a low-risk business opportunity and has become a booming industry across the globe. According to the Food and Agriculture Organization of the United Nations statistics from 2016, China[4] is the world's leading mushroom producer, while the United States and Italy are second and third. While the total production was more than 10 million ton in 2016[1].

2. History of Mushroom Farming

The initial cultivation technique of mushrooms was foraging, picking them from their natural habitats in the woods. During this period, mushrooms were thought to be mysterious fruits and were only accessible by the wealthy. Mushroom farming is believed to have started in 600 A.D.[2] in Asia with the discovery of mushrooms credited to the Chinese and Japanese. Mass production of the fungi in Asia was not established until the 16th century and it was not until the 19th century that this large scale mushroom farming operation found its way to Europe.

Fun Facts [2]**:** In the year around 1650, a Paris based melon discovered mushrooms in his fertilizer. Cultivating this exciting exclusive fruiting fungi, he introduce it to the Persian restaurants. It's when mushroom got a nickname "Parisian mushroom' in French. Later some farmers found the caves are the best place for cultivation and hence started growing mushrooms in the caves.

3. Types of Mushroom Farming

Mushrooms can be farmed using various methods. Besides gathering them straight from the woods, mushrooms can be grown using dead logs and mushroom plugs. Holes are drilled into the logs and then the plugs are plugged into the holes, each plug containing mushroom spores. Keeping the logs in damp areas will further enhance the growth of the mushrooms. Mushrooms can also be grown using plastic bags with small holes in them. A mixture of sawdust or hay and soil is placed into the plastic bag, along with mushroom spores. The bags are then hung and left in shaded or dark environment while the mushrooms

Fig 2: Wood Mushroom

Fig 3: Dead Logs Mushroom

Fig 4: Mushroom Cultivation in Plastic Bags

decompose the organic matter. After a week or so, mushrooms will begin to sprout through the small holes in the plastic bag.

4. Oyster Mushroom[3]

The oyster mushroom got its name from its strikingly similar resemblance to a fresh schucked oyster. Though looking like one particular mollusk, there happen to be three different types of oyster mushrooms: Pleurotus ostreatus, Pleurotus pulmonarius, and Pleurotus populinus. The only true difference between them is the season in which they grow. When in the wild, all can be found growing in a layered formation; the mushrooms stacked on top of each other in clusters pertruding from decaying wood.

Fig 6: Ground Oyster Mushroom

5. Required Materials for Cultivation

Production materials vary depending upon the scale of the farm. For this document, we will focus on small scale mushroom farming. The necessary materials are Cellulose waste, Straw, Water, Machete Straw grinder Boiling pot, Sterilizer, Temperature control device Sacks, Drums, Calcium carbonate(chalk)

6. Cultivation Process is broken down and described in several procedures below

6.1. Selection of Raw Materials

Straw is the base organic material for the production of oyster mushroom. It can be obtained from a variety of crops such as rice, wheat, barley, rye, sunflower, oats, buckwheat, peanuts hull etc. Straw should contain moisture levels of less than 12%[5], ensuring a predominantly dry material free of mold. Additives including limestone, gypsum, suger, green hay, alfalafa, wheat, and rice can be mixed into the straw to boost mushroom production. Sawdust from hardwoods can also be used in place of straw.

Fig 5(a): Good Quality Straw

Fig 5(b): Bad Quality Straw

6.2. Storage of Raw Materials

Raw materials should be stored out of the elements and kept dry to avoid mold formation and loss of materials. If it is not possible to store the raw materials in such an environment and they must be kept outside, covering them with a tarp is necessary to maintain a quality organic material.

6.3. Cutting the Raw Materials:

After the storing, chopping the straw is first step of the core processing. The collected straw(or raw materials) are cut into small pieces, usually the length of the cut is 5-10 cm and are soaked in fresh water for 8-16 hours. Straw are light and less distributed organic material. Even after boiling, their density does not increase much, thus cutting is needed to tightly pack the raw materials into the plastics bag. Better the density, better the growth of mushroom. Also different external materials such as sunflower husks, cottonseed hulls are also used to increase the density. In some parts of the world only husks or hulls are used to grow commercial mushroom but in general they are used with straw.

Fig 7: Farmers cutting straw for substrate using local method in Nepal

6.4. Preparation of the Substrate

Once the cutting process is completed and before the "heating treatment", straw is mixed with some minerals such as limestones, gypsum, suge are also added during the preparation of substrates. These minerals act as sterilization as well as fertilizers. In addition, wheat straw, wood chips, sawdust, sugar beet pulp etc can also be added as substrate. But it all depends on availability and are not the strict requirements. These materials should be chosen carefully as they may contaminate the whole mixture.

Standard Practices Used to Prepare Substrate

Agriculture straw such as corncob 53% or sawdust 53% or straw 53%, biogas residues or fungi residues 30%, rice bran or wheat bran 10%, peanut bran 2%, gypsum powder 1%, lime powder 2%, compound fertilizer 2%, the proportion between material and water is 1 to 1.25.

Fig 8: Preparation of Substrate

After the preparation of substrate, they are compressed into the bags and send for the heating treatment. Some growers pasteurize and sterile the substrate in a bulk before filling them to the bag. Usually, the first method is recommended as it prevents the contamination risk. Small hole of diameter 2-3 cm is made at the center the bag for inoculation.

Country	Dimensions (cm)	Material Used
Vietnam	20x30 or 18x25	Polyethene
India	45x30x15	Wooden Frame
South Africa	200x50	Polyethene
United States	23x13x74	Polyethene
Bangladesh	25x20	Polyethene
Nepal	36x61 or 46x66	Polyethene

Table 1: Packing bags sizes in different countries[5]

6.5. Heating Treatment

"It is process of killing the pests and microorganism by the method of pasteurization and sterilization"

6.5.1. Sterilization

Sterilization is process of killing microorganisms which are harmful to mycelium growth. Heating temperature depends on the quality of substrate. If the quality is good, serialization temperature is maintained in between 95 -199 degree Celsius and the substrate is heated for about 4-5 hours. But if the substrate contains sands or dirts, it should be heated in high temperature ranging from 200 - 600 degree Celsius depending upon the quantity of dirts or sands.

Fig 9: Vertical Partition Drum

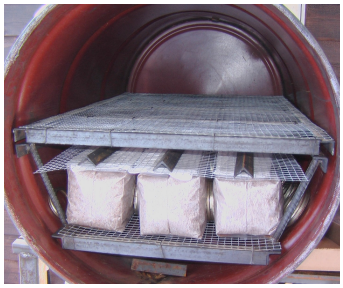

Fig 10: Horizontal Partition Drum

6.5.2. Pasteurization

For the small scale production, oil drums are used for the pasteurization. It is a process of heating the bas using steams. The drums are coated or formed a good compartment inside to avoid burning of bags. The drums in layers into two half vertically or 2-3 half horizontally, as shown into the figure below. The lower half of the drum is filled with water and upper half with bags. After placing the bags in the drums, lead is closed tight from the top to build sufficient pressure. The heating time depends upon the size of the bags, but usually they are heated for about 4-5 hours.

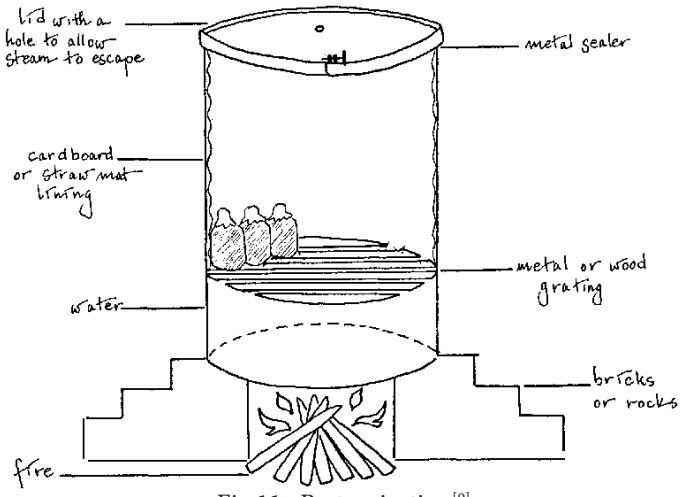

Fig 11: Pasteurization[9]

After the seaming, bags are cool down to 25 degree Celsius.

6.6. Spawning

Spawning is a process to transfer mushroom mycelia to the bags. These can be found on farmers shops or from commercial spawn supplier.

6.7. Inoculation

This is carried out in a clean room to avoid contamination and the people who perform this tasks use sterile gloves. About a tea spoonful of spawn is placed in the substrate using the hole made earlier on center of the bag.

In some places the conditioned straw is kept on the bags layer wise and on the periphery of each layer, spawn is sprayed. Generally, 250g bottle can be sprayed to 3 packs.

6.8. Incubation

After the inoculation bags are moved to dark incubation rooms where the temperature are optimized to 20-25 degree Celsius for the best results. They are kept in the incubation rooms for about 15-25 days, depending upon size, material and condition of the bags. These bags can be arranged in various fashion. Generally, they are placed in rack and the touching between the bags are avoided. One of the method is shown in the image below. As soon as the tiny mycelial fill up the bags, they are moved to the production room. In some case same room can be used as production. The room should have light for at least 10-12 hours for mycelial to grow into fruiting body.

Fig 12: Incubation Using Small Bags[10]

Small holes throughout the bags are made for the mycelial to develop into a fruiting body. Maintaining the atmospheric humidity, CO2, and water is very important. Mushroom contains about 90% of water thus maintaining humidity of the atmosphere is critically important during the early stage i.e. fruiting stage. Frequent watering about 2-3 buckets 5 to 6 times during the reproductive growing phase is required. Supplying water directly to the mushroom should be avoided.

Fig 13: Incubation Using Large Bags[11]

Some references [7] to environmental condition based on king oyster mushroom requirements is shown in the table below.

Parameters	Spawn phase	Pinhead	Fruiting body
Temperature (°C)	24–25	10–15	15–21
Relative humidty(%)	90–95	95–100	85–90
CO2 (ppm)	5000–20,000	500-1000	≤2000
Air replacement	1	4–8	4–5
Light (lux day−1)	-	500-1000	500-1000

Table 2: Environmental setup for different stages

7. Harvesting and Storage

The mushrooms should be ready to harvest in approximately 3 to 5 days after you see the first mushrooms begin to form. When the cap of the largest mushroom in the cluster begins to go from convex (turning down at the edges) to concave (flattening out or turning up at the edges), pick the whole cluster. Blue oyster and elm oyster mushrooms grow in clusters of multiple mushrooms, and it is generally best to pick the entire cluster when you see that the largest mushrooms in the cluster are ready to harvest. A white powder on the side of the bag means the mushrooms are simply

dropping their seeds. This is normal and safe. Harvest your mushrooms right away.

Fig 12: Harvest Ready Oyester Mushroom

8. Hygiene and some market statistics

Mushroom is widely considered as a power house of nutrition. It contains moderate amount of fibers and several minerals, vitamins(B, D), copper, iron, potassium, magnesium, and more interestingly it's low in calories and fat, 100% cholesterol free. It also contains some antioxidants which is believed to prevent cancer. The mushroom made up $1.22 billion[14] of revenue for the U.S. economy in 2016-17, with 929 million pounds being produced. By 2019, the world market is expected to produce $50,034.12 million[13].

9. Conclusion

The above description of mushroom farming will yield a quality product. There are other techniques used to achieve similar results, but this technique will be the most efficient way to turn a profit with a small scale operation as it uses locally available materials and with a minimal investment. This particular process also accounts for distributing a quality product that creates customer retention. Adhering to this guide will ensure you are well on your way to making a decent profit with minimal costs, while also establishing a growing clientele to turn your small-scale mushroom farm into wa mushroom farming empire.

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